Chapter 70 — Peer-to-Peer / Nearby Connections

> **Part VIII — Real-Time & Connectivity** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 69 (BLE), Ch 51 (permissions), Ch 46 (foreground services), Ch 61 (large files/streaming), Ch 2 (retries), Ch 44 (lifecycle), Ch 28 (privacy). Pairs with Ch 68 (network state), Ch 17 (uploads).
>
> **Reference surface in StreamCart:** offline, device-to-device features — sharing a product/receipt or a large media file directly between two nearby phones, a cashier handoff, a "nearby" multiplayer or group action — using **Nearby Connections** (or Wi-Fi Direct) so data moves **peer-to-peer without the internet**.
>
> **Note:** this chapter completes **Part VIII — Real-Time & Connectivity (Ch 63–70)**.

70.1 Problem Definition and Real-World Context

Sometimes two devices need to exchange data **directly, nearby, without a server or the internet**: sharing a large file at a trade show with no Wi-Fi, a cashier handing off a cart to a customer's phone, a quick "send to the person next to me," or local multiplayer. Routing this through the backend is wasteful (or impossible offline) and slow for big payloads. The platform answer is **peer-to-peer (P2P)** connectivity — primarily Google's **Nearby Connections API** (which transparently uses Bluetooth, BLE, and Wi-Fi under the hood), or lower-level **Wi-Fi Direct (Wi-Fi P2P)** — to **discover, connect, and transfer** between nearby devices directly.

Concretely, teams hit:

- **Choosing the mechanism. Nearby Connections** (high-level, auto-selects radios, handles discovery/connection/transfer) vs **Wi-Fi Direct** (lower-level, more control, more work) vs just **BLE** (Ch 69, tiny payloads). Picking wrong means reinventing what Nearby gives you, or using a heavyweight API for a trivial need.
- **Discovery + connection model.** Advertise/discover, request/accept a connection with a **mutual authentication token** (both sides verify the same code), then exchange payloads — a specific handshake to get right.
- **Payload types & large transfers.** Nearby supports **BYTES** (small), **FILE**, and **STREAM** payloads; large files must be **streamed** (Ch 61) with **progress** and **resume**, not loaded into memory.
- **Permissions (Ch 51).** Nearby needs a combination of **Bluetooth** (`BLUETOOTH_SCAN` / `ADVERTISE` / `CONNECT`, Ch 69), **Wi-Fi/** `NEARBY_WIFI_DEVICES` (Android 13+), and historically **location** — a matrix much like BLE (Ch 69).
- **Topology & reliability.** Point-to-point vs star vs cluster strategies; connections drop (range/interference) and need **retry** (Ch 2); transfers must survive and resume.
- **Security & trust.** No server mediates, so **authenticate the peer** (the mutual token), treat received data as **untrusted** (Ch 23), and get **user consent** — P2P is a physical-proximity trust model.
- **Lifecycle/battery (Ch 44/46).** Discovery/advertising and active transfers drain battery; scope to the session and use a **foreground service** (Ch 46) for ongoing transfers.

The architecture: prefer **Nearby Connections** for P2P (it auto-selects Bluetooth/BLE/ Wi-Fi and handles discovery/connection/transfer) over hand-rolling Wi-Fi Direct; implement the **advertise/discover → authenticated connect (mutual token) → payload exchange** handshake; **stream large payloads** (FILE/STREAM) with progress/resume (Ch 61); manage the **permission matrix** (Bluetooth + Wi-Fi/nearby + legacy location, Ch 51/69); choose a **strategy/topology** (P2P_POINT_TO_POINT/STAR/CLUSTER); handle **drops with retry** (Ch 2) and **scope to lifecycle/FGS** (Ch 44/46); and **secure** it — authenticate the peer, untrusted data, consent (Ch 23/28). This chapter is about **direct device-to-device data transfer without the internet**.

Three practical use cases

1. **Share a large media file (StreamCart).** Two nearby phones transfer a big video offline via a Nearby **FILE** payload, streamed with progress (Ch 61) — no server, no data plan.
2. **Cashier → customer cart handoff.** A store device advertises; the customer's phone discovers and connects (authenticated token shown on both), receives the cart as a BYTES payload.
3. **Nearby group action / local multiplayer.** A STAR topology where one host connects to several nearby devices for a shared local session.

Why this is architectural

P2P/Nearby composes **radio selection (BLE/Wi-Fi, Ch 69/68)**, **permissions (Ch 51)**, **foreground services (Ch 46)**, **streaming (Ch 61)**, **retries (Ch 2)**, and a **proximity trust model (Ch 23/28)** — a distinct connectivity paradigm (no server). The decisions — Nearby vs Wi-Fi Direct vs BLE, authenticated handshake, payload streaming, permission matrix, topology, security — determine whether offline device-to-device transfer is reliable and safe.

70.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Discover and connect to nearby devices directly (no internet).
F2	Authenticate the connection (mutual token both sides verify).
F3	Exchange payloads: small (BYTES) and large (FILE/STREAM, streamed + progress/resume) (Ch 61).
F4	Handle the permission matrix (Bluetooth + Wi-Fi/nearby + legacy location) (Ch 51/69).
F5	Choose topology/strategy; handle drops with retry; scope to lifecycle/FGS (Ch 2/44/46).
F6	Secure: authenticate peer, untrusted data, user consent (Ch 23/28).

Non-functional requirements

Attribute	Policy
Offline	Works with no internet (direct radios).
Throughput	Large files stream efficiently (Wi-Fi radio); progress/resume (Ch 61).
Reliability	Retry drops; resume transfers (Ch 2).
Battery	Discovery/advertising/transfer scoped to session; FGS for ongoing (Ch 44/46).
Security	Peer authenticated; data untrusted; consent (proximity trust, Ch 23/28).

Scope

- **In scope:** Nearby Connections (strategies/topologies, advertise/discover, authenticated connect, payload types BYTES/FILE/STREAM, streaming large + progress), Wi-Fi Direct (overview/when), BLE for tiny payloads (contrast Ch 69), permission matrix (Ch 51/69), retry/reconnect (Ch 2), lifecycle + FGS (Ch 44/46), security (peer auth, untrusted data, consent, Ch 23/28).
- **Out of scope:** BLE GATT detail (Ch 69), streaming internals (Ch 61), permission state machine (Ch 51), FGS mechanics (Ch 46), network state (Ch 68), server-mediated transfer (Ch 17).

Assumptions & constraints

- **Nearby Connections API** (Google Play services) as primary; Wi-Fi Direct/BLE as alternatives; `minSdk 24`, `targetSdk` current (Android 13+ `NEARBY_WIFI_DEVICES`). P2P is **physical-proximity** trust with **no server**; large payloads need streaming (Ch 61); radios drop.

70.3 System Design Fundamentals

- **Prefer Nearby Connections (high-level) over raw Wi-Fi Direct/BLE.** Google's **Nearby Connections API** abstracts P2P: it **auto-selects** the best radio (Bluetooth, BLE for discovery, **Wi-Fi** for high-throughput transfer), and handles **discovery, connection, authentication, and payload transfer** with a clean callback API. Hand-rolling **Wi-Fi Direct (Wi-Fi P2P)** gives more control (and works without Play services) but is far more work (group formation, sockets, IP). Use **BLE** (Ch 69) only for **tiny** payloads. Default to **Nearby Connections**.
- **Strategies / topology.** Nearby offers **strategies**:
  - **`P2P_POINT_TO_POINT`:** one-to-one, highest bandwidth (large file transfer).
  - **`P2P_STAR`:** one host ↔ many clients (group/host session).
  - **`P2P_CLUSTER`:** many-to-many mesh (flexible, lower bandwidth). Choose by use case (file share → point-to-point; local multiplayer → star/cluster).
- **The handshake: advertise/discover → authenticate → connect → transfer.**
  1. One device **advertises** (`startAdvertising`), the other **discovers** (`startDiscovery`) by a shared **service id**.
  2. On finding, one **requests a connection**; both sides receive an **authentication token** and must **verify it matches** (shown to users / auto-verified) — this establishes **mutual trust** (anti-MITM for the pairing).
  3. On both **accepting**, the connection is established; then **payloads** flow.

- **Payload types (small vs large):**
  - **BYTES:** small (≤ a few KB) — a cart, a message, metadata.
  - **FILE:** a file — Nearby transfers it efficiently (over Wi-Fi), with **progress** callbacks.
  - **STREAM:** a continuous stream (e.g., audio/sensor). Large files must be handled as **FILE/STREAM streamed** (Ch 61) — never loaded wholly into memory — with **progress** and ideally **resume** on reconnect (Ch 2/61).
- **Permission matrix (Ch 51/69).** Nearby needs a combination:
  - **Bluetooth:** `BLUETOOTH_SCAN` / `BLUETOOTH_ADVERTISE` / `BLUETOOTH_CONNECT` (Android 12+, Ch 69).
  - **Wi-Fi:** **`NEARBY_WIFI_DEVICES`** (Android 13+, with `neverForLocation` if not deriving location), else historically **`ACCESS_FINE_LOCATION`** (Ch 67). Much like BLE's matrix (Ch 69); request **in context** with graceful denial (Ch 51).
- **Reliability: drops + retry + resume (Ch 2).** P2P links drop (range, interference, device sleep). Handle **disconnection** (`onDisconnected`), **retry** connection with backoff (Ch 2), and **resume large transfers** (track bytes transferred; re-send the remainder, Ch 61) so a dropped multi-GB transfer doesn't restart from zero.
- **Lifecycle & battery (Ch 44/46).** Discovery/advertising and active transfers are battery-heavy and should be **scoped to the session** (Ch 44): advertise/discover only while the share UI is open; **stop promptly**. For a **long transfer**, use a **foreground service** (Ch 46, `connectedDevice` / `dataSync`) with progress so it survives backgrounding.
- **Security: proximity trust + authenticate + untrusted (Ch 23/28).** There's **no server** mediating, so:
  - **Authenticate the peer:** verify the **mutual authentication token** (users confirm the same code, or app-layer verification) — don't accept connections blindly.
  - **Treat received data as untrusted** (Ch 23): validate payloads; a nearby device could be malicious (sending malware, spoofing).
  - **User consent:** both users explicitly accept the connection/transfer; show what's being shared.
  - Device identity/proximity is **sensitive** (Ch 28) — minimize, don't log.

Core insight: **for offline device-to-device transfer, prefer the Nearby Connections API — it auto-selects Bluetooth/BLE/Wi-Fi and handles discovery, authenticated connection, and payload transfer — over hand-rolling Wi-Fi Direct (more control, more work) or using BLE (Ch 69, tiny payloads only); run the advertise/discover → mutual-token-authenticated connect → payload-exchange handshake, choosing a strategy (point-to-point for big files, star/cluster for groups); stream large FILE/STREAM payloads with progress and resume (Ch 61); navigate the Bluetooth + Wi-Fi(`NEARBY_WIFI_DEVICES`)/legacy-location permission matrix (Ch 51/69); handle drops with retry/backoff and resume (Ch 2), scoped to the active lifecycle with a foreground service for long transfers (Ch 44/46); and secure it as a physical-proximity trust model — authenticate the peer, treat received data as untrusted, require explicit consent (Ch 23/28).**

70.4 High-Level Architecture


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Components & ownership

- **Mechanism choice:** Nearby (default) vs Wi-Fi Direct vs BLE.
- **Permission gate** (Ch 51/69) → **authenticated handshake** (mutual token) → **payload exchange** (BYTES small / FILE-STREAM streamed, Ch 61).

- **FGS** for long transfers (Ch 46); **retry/resume** on drop (Ch 2/61); **lifecycle** scopes discovery; **security** (peer auth, untrusted, consent).

Direction: need → Nearby → permissions → authenticated connect → payload (stream large) → FGS/retry/resume; session-scoped; secured by proximity trust.

70.5 Detailed Component Design



Component	Responsibility
`NearbyManager`	advertise/discover/connect/send via Nearby Connections
`ConnectionLifecycle`	authenticate (token), handle result/disconnect + retry (Ch 2)
`PayloadHandler`	BYTES (small) + FILE/STREAM (streamed + progress, Ch 61)
`Strategy`	point-to-point / star / cluster topology

70.6 End-to-End Data Flow

Authenticated connect + file transfer



Mechanism choice



70.7 Complete Kotlin Implementation

Package: `com.streamcart.nearby`.

70.7.1 Permissions (Nearby matrix, Ch 51/69)

```
<!-- Bluetooth (Ch 69) -->
<uses-permission android:name="android.permission.BLUETOOTH_SCAN"
android:usesPermissionFlags="neverForLocation" />
<uses-permission android:name="android.permission.BLUETOOTH_ADVERTISE" />
<uses-permission android:name="android.permission.BLUETOOTH_CONNECT" />
<!-- Wi-Fi (Android 13+) -->
<uses-permission android:name="android.permission.NEARBY_WIFI_DEVICES"
android:usesPermissionFlags="neverForLocation" />
<!-- Legacy (pre-13) Wi-Fi scan needed location (Ch 67) -->
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"
android:maxSdkVersion="32" />
<uses-permission android:name="android.permission.FOREGROUND_SERVICE_CONNECTED_DEVICE" />  <!--
Ch 46 -->
```

70.7.2 NearbyManager: advertise/discover + authenticated connect

```
@Singleton
class NearbyManager @Inject constructor(@ApplicationContext private val ctx: Context) {
    private val client = Nearby.getConnectionsClient(ctx)
    private val strategy = Strategy.P2P_POINT_TO_POINT     // highest bandwidth for file share

    fun advertise(name: String) = client.startAdvertising(
        name, SERVICE_ID, connectionLifecycle,
AdvertisingOptions.Builder().setStrategy(strategy).build())

    fun discover(): Flow<DiscoveredEndpoint> = callbackFlow {
        val cb = object : EndpointDiscoveryCallback() {
            override fun onEndpointFound(id: String, info: DiscoveredEndpointInfo) =
                trySend(DiscoveredEndpoint(id, info.endpointName)).let {}
            override fun onEndpointLost(id: String) {}
        }
        client.startDiscovery(SERVICE_ID, cb,
DiscoveryOptions.Builder().setStrategy(strategy).build())
        awaitClose { client.stopDiscovery() }              // stop promptly (battery, Ch 44)
    }

    fun requestConnection(name: String, endpointId: String) =
        client.requestConnection(name, endpointId, connectionLifecycle)

    private val connectionLifecycle = object : ConnectionLifecycleCallback() {
        override fun onConnectionInitiated(endpointId: String, info: ConnectionInfo) {
            // AUTHENTICATE: verify the SAME token on both sides (show to user / app-layer) —
anti-MITM.
            val token = info.authenticationDigits          // both devices see the same digits
            // ... present token for user confirmation, then:
            client.acceptConnection(endpointId, payloadCallback)   // only after mutual
verification/consent
        }
        override fun onConnectionResult(endpointId: String, result: ConnectionResolution) {
            if (result.status.statusCode == ConnectionsStatusCodes.STATUS_OK) { /* connected */
}
        }
        override fun onDisconnected(endpointId: String) { /* retry + resume (Ch 2/61) */ }
    }
}
```

70.7.3 Payloads: BYTES (small) + FILE streamed with progress (Ch 61)

```
private val payloadCallback = object : PayloadCallback() {
    override fun onPayloadReceived(endpointId: String, payload: Payload) {
        when (payload.type) {
            Payload.Type.BYTES -> handleBytes(payload.asBytes()!!)           // small
(untrusted! Ch 23)
            Payload.Type.FILE -> {
                val uri = payload.asFile()?.asUri()                          // large -> stream,
don't load (Ch 61)
                // copy/stream from uri to app storage via ContentResolver (Ch 61/62)
            }
            Payload.Type.STREAM -> handleStream(payload.asStream())          // continuous
        }
    }
    override fun onPayloadTransferUpdate(endpointId: String, update: PayloadTransferUpdate) {
        val progress = update.bytesTransferred.toFloat() / update.totalBytes  // progress (Ch
61)
        // update UI; on FAILURE -> retry + resume remainder (Ch 2/61)
    }
}

// Send a large file as a FILE payload (Nearby streams it over Wi-Fi):
fun sendFile(endpointId: String, file: File) =
    client.sendPayload(endpointId, Payload.fromFile(file))   // streamed; progress via
onPayloadTransferUpdate
```

70.7.4 Long transfer in a foreground service (Ch 46)

```
// A multi-GB transfer runs inside a connectedDevice/dataSync FGS (Ch 46) with a progress
notification,
// so it survives backgrounding; stop the FGS + stopAll endpoints when the transfer completes.
@AndroidEntryPoint
class NearbyTransferService : Service() {
    @Inject lateinit var nearby: NearbyManager
    override fun onStartCommand(intent: Intent?, flags: Int, startId: Int): Int {
        startForeground(NOTIF_ID, transferNotification(0),
ServiceInfo.FOREGROUND_SERVICE_TYPE_CONNECTED_DEVICE)
        // run transfer; update notification from onPayloadTransferUpdate; stopSelf on complete
        return START_NOT_STICKY
    }
    override fun onBind(i: Intent?): IBinder? = null
}
```

70.7.5 Lifecycle + stop promptly

```
// Scope discovery/advertising to the share session (Ch 44); stop when done (battery).
fun stopSession() {
    client.stopAdvertising(); client.stopDiscovery(); client.stopAllEndpoints()   // release
radios
}
```

70.7.6 Module structure

```
:feature:nearby  NearbyManager, ConnectionLifecycle, PayloadHandler, NearbyTransferService (FGS,
Ch 46)
# Permissions -> Ch 51/69; streaming large payloads -> Ch 61/62; retries -> Ch 2; lifecycle ->
Ch 44.
```

70.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Accept before token verified	MITM/unauthorized	verify mutual token + consent before `acceptConnection`
Large payload loaded into memory	OOM	FILE/STREAM streamed (Ch 61), not BYTES
Transfer drops mid-file	restart from 0	track bytes; retry + resume remainder (Ch 2/61)
Discovery/advertising left on	battery drain	stop promptly; session-scoped (Ch 44)
Multiple simultaneous connections	contention	choose strategy (point-to-point limits to one)
Received data trusted	malware/spoof	validate untrusted payloads (Ch 23)
Permission missing	discovery/connect fails	permission matrix (Ch 51/69)

- **Authenticate before accepting:** `acceptConnection` must come **after** both sides verify the **same authentication token** and the users consent — accepting blindly allows a MITM or unwanted peer (the proximity trust model's core safeguard).
- **Stream large payloads, don't BYTES them:** a big file as a BYTES payload OOMs; use **FILE/STREAM** which Nearby transfers efficiently (over Wi-Fi) and stream it to storage (Ch 61/62).
- **Resume, don't restart:** track `bytesTransferred` and on a drop resume the remainder (Ch 2/61) so a large transfer survives interruptions.

70.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Connection drops mid-transfer	`onDisconnected`/FAILURE	retry + resume remainder (Ch 2/61)	completes
OOM on large payload	crash	FILE/STREAM streamed (Ch 61)	safe
MITM / wrong peer	token mismatch	verify mutual token + consent before accept	blocked
Discovery finds nothing	no endpoints	fix permission matrix (Ch 51/69); same service id	connects
Battery drain	radios left on	stop advertising/discovery/endpoints promptly (Ch 44)	battery ok
No Play services	Nearby unavailable	Wi-Fi Direct fallback	works
Malicious data	validation	treat received payloads as untrusted (Ch 23)	safe



> **Nearby does the hard radio work; your job is the handshake, streaming, and trust.** Authenticate the peer with the mutual token before accepting, stream large payloads (FILE/STREAM) with progress and resume, get the permission matrix right, scope radios to the session (FGS for long transfers), and validate received data as untrusted — then offline device-to-device transfer is reliable and safe.

70.10 Scaling the Implementation

- **Level 1.** Nearby advertise/discover + authenticated connect (verify token) + BYTES payload.
- **Level 2.** FILE/STREAM payloads streamed with progress (Ch 61); permission matrix (Ch 51/69); stop radios promptly.
- **Level 3 — production.** Retry/backoff + resume on drop (Ch 2/61); `connectedDevice` FGS for long transfers (Ch 46); strategy selection; consent + peer auth (Ch 23).
- **Level 4 — large.** STAR/CLUSTER group sessions; multi-file transfers; Wi-Fi Direct fallback (no Play services); transfer analytics.
- **Level 5 — enterprise.** P2P governance, offline-mesh scenarios, secure provisioning, cross-platform (iOS via a common protocol), throughput SLOs.

Avoid overengineering: for most "share nearby" needs, **Nearby Connections + authenticated connect + FILE payload streamed** is enough. Don't hand-roll Wi-Fi Direct unless you lack Play services or need low-level control; don't use BYTES for large files.

70.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Mechanism	Nearby Connections	Wi-Fi Direct	Nearby (auto-radio, handles handshake/transfer); Wi-Fi Direct for low-level / no Play services
Mechanism	Nearby Connections	BLE (Ch 69)	Nearby for real transfers; BLE for tiny payloads only
Payload (large)	FILE/STREAM streamed (Ch 61)	BYTES	FILE/STREAM (BYTES OOMs for large)
Strategy	Point-to-point (file) / star (group)	one for all	Match to use case
Transfer	Via P2P (offline, fast local)	via backend (Ch 17)	P2P when offline/local/large; backend when server needs it
Drop handling	Retry + resume (Ch 2/61)	restart	Resume (large transfers)
Trust	Authenticated token + consent	accept blindly	Authenticate (proximity trust, Ch 23)

Nearby Connections vs Wi-Fi Direct: Nearby auto-selects the best radio (BLE for discovery, Wi-Fi for throughput), handles discovery/authentication/transfer with a clean API, and "just works" — but needs **Play services**. **Wi-Fi Direct (Wi-Fi P2P)** is lower-level (you manage group formation, sockets, IP), more work, but available without Play services and offers more control. Default to **Nearby**; use Wi-Fi Direct only when you lack Play services or need that control.

P2P vs backend transfer (Ch 17): routing a large nearby transfer through the **backend** (upload then download, Ch 17) requires internet, burns data, and is slow for big files between co-located devices. **P2P** moves it **directly** (offline, over the local Wi-Fi radio) — ideal when devices are nearby and especially when offline. Use P2P for local/offline/large transfers; the backend when the server must see/process the data.

FILE/STREAM vs BYTES for large payloads: a BYTES payload holds everything in memory (OOM for large files); **FILE**/**STREAM** payloads let Nearby transfer efficiently over Wi-Fi with progress and let you **stream** to storage (Ch 61). Use BYTES only for small metadata; FILE/STREAM for anything large.

70.12 Security & Privacy Considerations

- **Proximity trust + authenticate the peer (Ch 23):** no server mediates, so verify the **mutual authentication token** (both devices show the same digits; users confirm, or app-layer verification) **before accepting** — this is the anti-MITM safeguard. Never accept connections blindly.
- **Treat all received data as untrusted (Ch 23):** a nearby device could send malware, spoofed content, or malformed payloads — validate/sanitize everything received; don't auto-execute or trust a received file.
- **Explicit consent (Ch 28):** both users explicitly accept the connection and the transfer; show **what** is being shared and with **whom** — surprise P2P transfers are a trust/abuse risk.
- **Least permission (Ch 51/69):** request Bluetooth + `NEARBY_WIFI_DEVICES` (with `neverForLocation` if not deriving location) in context; avoid legacy location where possible.

- **Device identity is sensitive (Ch 28):** endpoint names/ids can identify users — minimize, don't log; use non-identifying advertise names.
- **Encrypt sensitive payloads:** Nearby encrypts the link, but for highly sensitive data, add app-layer encryption (Ch 25) and authenticate at the app layer.
- **Stop radios promptly:** leaving advertising/discovery on is a discoverability/privacy concern as well as battery.
- Client vs backend: P2P is serverless; there's **no backend to authorize** the transfer, so the **peer authentication + user consent + untrusted-data handling** carry the entire trust burden — design them carefully.

70.13 Performance and Resource Optimization

- **Nearby uses Wi-Fi for throughput** automatically — large files transfer fast locally (far faster than cellular upload/download).
- **Stop advertising/discovery/endpoints promptly** (Ch 44) — the top battery lever; radios on = drain.
- **Stream large payloads** (FILE/STREAM, Ch 61) — no OOM; progress-driven.
- **Resume, don't restart** (Ch 2/61) — don't re-transfer gigabytes on a drop.
- **Choose strategy by need** — point-to-point maximizes a single transfer's bandwidth.
- **FGS for long transfers** (Ch 46) so they complete backgrounded without the user holding the screen.
- **Profiling:** transfer throughput, connect success rate, drop/resume rate, radio battery, discovery time.

70.14 Testing Strategy

Nearby is hard to unit-test (needs two real devices + Play services); test the **handshake/ payload/resume logic** in isolation with a fake client, and do **two-device instrumentation**.

```
class ConnectionAuthTest {
    @Test fun `does not accept before token verified`() {
        lifecycle.onConnectionInitiated("E1", connectionInfo(token = "1234"))
        assertFalse(fakeClient.accepted("E1"))              // waits for verification/consent
        lifecycle.userConfirmsToken("E1", "1234")
        assertTrue(fakeClient.accepted("E1"))
    }
    @Test fun `rejects on token mismatch`() { /* user sees different digits -> reject */ }
}

class PayloadTest {
    @Test fun `large file sent as FILE streamed, not BYTES`() { /* assert Payload.Type.FILE */ }
    @Test fun `resumes remainder after drop`() = runTest { /* bytesTransferred tracked -> resume
(Ch 61) */ }
    @Test fun `received payload validated as untrusted`() { /* malformed -> rejected (Ch 23) */
}
}
```

What to test: authenticated connect (accept only after token verify + consent; reject on mismatch); large payload uses FILE/STREAM (not BYTES); resume after drop (Ch 2/61); untrusted-payload validation (Ch 23); permission-gated discovery (Ch 51/69); stop-radios. Fake the Nearby client for logic; **two real devices with Play services** for end-to-end.

70.15 Production Observability

- **Metrics:** connect success rate; **transfer throughput** + completion rate; drop/resume rate; discovery time; radio battery; permission grant rate (Ch 51/69); rejected-connection (token-mismatch) count; Play-services-unavailable rate (→ Wi-Fi Direct fallback).
- **Logs:** `nearby{event, strategy, payload_type, bytes, result}` — **no endpoint names/ device identity/PII** (Ch 28).
- **Debugging:** no discovery → permission matrix (Ch 51/69) / mismatched service id; OOM → BYTES for large (use FILE, Ch 61); transfer restarts → no resume (Ch 2/61); battery → radios not stopped (Ch 44); MITM concern → accepting before token verify; Play-services missing → fallback.
- **Alerting:** connect-success drop; transfer-completion drop; rejected-connection spike (possible abuse).
- **Dashboards:** connect/transfer success + throughput; drop/resume rate; permission funnel.

70.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *What is P2P/Nearby for?* Offline, direct device-to-device transfer (no server/internet).
2. *Primary API?* Nearby Connections (auto-selects BT/BLE/Wi-Fi).
3. *Nearby vs Wi-Fi Direct?* High-level auto-radio vs low-level control / no Play services.
4. *Nearby vs BLE (Ch 69)?* Real transfers vs tiny payloads.
5. *Radios used?* BLE for discovery, Wi-Fi for throughput (Nearby picks).

Handshake 6. *Steps?* Advertise/discover → request connection → verify mutual token → accept → transfer. 7. *Authentication?* Both sides verify the same token (anti-MITM) + consent. 8. *Service id?* Shared id both sides advertise/discover on.

Payloads 9. *Types?* BYTES (small), FILE (file), STREAM (continuous). 10. *Large files?* FILE/STREAM streamed (Ch 61) — never BYTES (OOM). 11. *Progress?* `onPayloadTransferUpdate`. 12. *Drop mid-file?* Resume remainder (Ch 2/61).

Strategy 13. *Strategies?* P2P_POINT_TO_POINT / STAR / CLUSTER. 14. *File share?* Point-to-point (bandwidth). 15. *Group/multiplayer?* Star/cluster.

Permissions (Ch 51/69) 16. *Matrix?* Bluetooth (`SCAN` / `ADVERTISE` / `CONNECT`) + `NEARBY_WIFI_DEVICES` (13+) / legacy location. 17. *`neverForLocation`?* If not deriving location (Ch 67).

Lifecycle/battery 18. *Scope?* Session; stop advertising/discovery/endpoints promptly (Ch 44). 19. *Long transfer?* `connectedDevice` FGS (Ch 46).

Security (Ch 23/28) 20. *Trust model?* Physical proximity + authenticate peer + consent (no server). 21. *Received data?* Untrusted — validate. 22. *Device identity?* Sensitive — minimize/don't log. 23. *Sensitive payloads?* App-layer encryption (Ch 25).

Testing/Production 24. *Test end-to-end?* Two real devices + Play services. 25. *No discovery?* Permission matrix / service id. 26. *OOM?* BYTES for large (use FILE).

Design-change challenges

1. **Share a large video between two phones offline.** Nearby point-to-point + FILE payload streamed + progress (Ch 61).
2. **Prevent connecting to the wrong/malicious device.** Verify mutual token + consent before accept (Ch 23).
3. **Transfer restarts from 0 on a drop.** Track bytes; resume remainder (Ch 2/61).
4. **No discovery on Android 13.** Add `NEARBY_WIFI_DEVICES` (Ch 51).
5. **No Play services on the device.** Wi-Fi Direct fallback.

Mock walkthrough (abridged)

Clarify (offline?, payload size, group or 1:1, security?) → Nearby Connections (vs Wi-Fi Direct/BLE) → strategy (point-to-point/star) → permission matrix (Ch 51/69) → handshake (advertise/discover → verify token → accept) → payloads (BYTES small / FILE-STREAM streamed

- progress + resume, Ch 61/2) → FGS for long transfer + stop radios (Ch 46/44) → security (authenticate peer, untrusted data, consent, Ch 23/28) → observability.

70.17 Practical Exercises

1. **Beginner.** Advertise/discover with Nearby on a shared service id and exchange a small BYTES payload after verifying the mutual token. *Eval:* connects; token-verified.
2. **Intermediate.** Send a large file as a FILE payload, streamed with a progress bar (not BYTES); handle the permission matrix (Ch 51/69). *Eval:* no OOM; progress; permission- correct.
3. **Advanced.** Add retry + resume on disconnect (resume the remaining bytes) and run the transfer in a `connectedDevice` FGS (Ch 46). *Eval:* survives drops/background.
4. **Production debugging.** Fix "no discovery on Android 13" (missing `NEARBY_WIFI_DEVICES`) and an OOM (BYTES for a large file → FILE). *Eval:* discovers; streams.
5. **System design challenge.** Secure the pairing (mutual token + consent + untrusted-data validation) and add a Wi-Fi Direct fallback for no-Play-services devices. *Eval:* no blind accept; works without Play services.

(Solutions withheld unless requested.)

70.18 Chapter Summary & Reference Sheet

Key decisions: for **offline, direct device-to-device transfer**, prefer the **Nearby Connections API** — it **auto-selects Bluetooth/BLE/Wi-Fi** and handles **discovery, authenticated connection, and payload transfer** — over hand-rolling **Wi-Fi Direct** (more control, more work, but no Play services needed) or using **BLE** (Ch 69, tiny payloads only); run the **advertise/discover → mutual-token-authenticated connect (verify the same token + consent before `acceptConnection`) → payload exchange** handshake, choosing a **strategy** (point-to-point for big files, star/cluster for groups); send small data as **BYTES** and **large files as FILE/STREAM payloads streamed** with **progress and resume** (never BYTES — OOM; Ch 61/2); navigate the **Bluetooth + `NEARBY_WIFI_DEVICES` (13+)/legacy- location permission matrix** (Ch 51/69); handle **drops with retry/backoff and resume the remainder** (Ch 2/61); **scope radios to the active session and stop promptly**, using a **`connectedDevice` foreground service** for long transfers (Ch 44/46); and secure it as a **physical-proximity trust model with no server — authenticate the peer, treat received data as untrusted, require explicit consent**, and minimize device identity (Ch 23/28).

Concepts: Nearby Connections (strategies/topologies, auto-radio), advertise/discover → authenticated connect (mutual token) → payload (BYTES/FILE/STREAM), streaming large payloads

- progress + resume, Wi-Fi Direct/BLE alternatives, permission matrix, lifecycle + FGS, proximity trust (authenticate/untrusted/consent).

Common mistakes: hand-rolling Wi-Fi Direct when Nearby suffices; BYTES for large files (OOM); accepting connections without verifying the mutual token (MITM); no resume (restart on drop); wrong permission matrix (no discovery); leaving radios on (battery); trusting received data; no consent; logging device identity.

Checklist: ☐ Nearby Connections (default) vs Wi-Fi Direct / BLE ☐ strategy per use case (point-to-point/star/cluster) ☐ advertise/discover → verify mutual token + consent → accept ☐ BYTES small / FILE-STREAM streamed + progress + resume (Ch 61/2) ☐ permission matrix (Bluetooth + `NEARBY_WIFI_DEVICES`/legacy location, Ch 51/69) ☐ retry + resume on drop (Ch 2/61) ☐ session-scoped radios + stop promptly + FGS for long transfer (Ch 44/46) ☐ authenticate peer + untrusted data + consent (Ch 23/28) ☐ device identity minimized/not logged ☐ Wi-Fi Direct fallback (no Play services) ☐ connect/transfer telemetry.

Key APIs: `Nearby.getConnectionsClient`, `startAdvertising` / `startDiscovery` / `stopAdvertising` / `stopDiscovery` / `stopAllEndpoints`, `requestConnection` / `acceptConnection`, `ConnectionLifecycleCallback` (`onConnectionInitiated` / `onConnectionResult` / `onDisconnected`), `ConnectionInfo.authenticationDigits`, `Payload.fromFile` / `fromBytes` / `fromStream` / `Payload.Type`, `PayloadCallback` (`onPayloadReceived` / `onPayloadTransferUpdate`), `Strategy` (`P2P_POINT_TO_POINT` / `STAR` / `CLUSTER`), `NEARBY_WIFI_DEVICES` /Bluetooth perms (Ch 51/69), `connectedDevice` FGS (Ch 46), Wi-Fi Direct (`WifiP2pManager`).

References:

- Nearby Connections: https://developers.google.com/nearby/connections/overview
- Nearby Connections strategies: https://developers.google.com/nearby/connections/strategies
- Wi-Fi Direct (Wi-Fi P2P): https://developer.android.com/develop/connectivity/wifi/wifi-direct
- NEARBY_WIFI_DEVICES: https://developer.android.com/develop/connectivity/wifi/wifi-permissions
- (composes) Ch 69 BLE, Ch 51 Permissions, Ch 46 Foreground Services, Ch 61 Large Files, Ch 2 Retries, Ch 44 Lifecycle, Ch 23/28 Security/Privacy

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/flow/failure diagrams; NearbyManager (advertise/discover/auth connect) + payload handler (BYTES/FILE streamed) + FGS + stop-radios code; concurrency (auth-before-accept, stream-large, resume); failure (Nearby-does-radio-you-do-handshake/trust principle); 5 levels; Nearby-vs-Wi-Fi-Direct, P2P-vs-backend, FILE-vs-BYTES trade-offs; security (proximity trust, authenticate, untrusted, consent); perf (Wi-Fi throughput, stop radios, resume); fake-client + two-device tests; observability (connect/transfer success); 26 Q&A + 5 challenges. ✅ Completes Part VIII.
- **Expand later:** BLE (Ch 69), large files (Ch 61), permissions (Ch 51), FGS (Ch 46).
- **Interview coverage:** exceeds ≥25 bar.